

6. Write a C program to define two functions to calculate the area of triangle & rectangle, using call by reference method.

```
#include <stdio.h>
```

```
// Function Prototypes using pointers for call by reference
```

```
void areaTriangle(float b, float h, float *a);
```

```
void areaRectangle(float l, float w, float *a);
```

```
int main(void)
```

```
{
```

```
    float base, height, length, width, area;
```

```
    // Triangle calculation
```

```
    printf("Enter base and height of triangle: ");
```

```
    scanf("%f %f", &base, &height);
```

```
    // Pass address of 'area' using &
```

```
areaTriangle(base, height, &area);  
printf("Area of Triangle = %.2f\n", area);
```

```
// Rectangle calculation  
printf("\nEnter length and width of rectangle: ");  
scanf("%f %f", &length, &width);
```

```
// Pass address of 'area' using &  
areaRectangle(length, width, &area);  
printf("Area of Rectangle = %.2f\n", area);
```

```
return 0;  
}
```

```
// Function to calculate triangle area
```

```
void areaTriangle(float b, float h, float *a)  
{  
    // Formula:  $0.5 * \text{base} * \text{height}$   
    *a =  $0.5 * b * h$ ;  
}
```

```
// Function to calculate rectangle area
```

```
void areaRectangle(float l, float w, float *a)
```

```
{  
  // Formula: length * width  
  *a = l * w;  
}
```